FREEDOM INTERNATIONAL SCHOOL

WORKSHEET-MCQ (SOLUTIONS)

PHYSICS

CLASS XII

WAVE OPTICS

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1. In Young's double slit experiment, the separation between the slits is halved and distance between the slits and screen is doubled. The fringe width is
   (a) unchanged (b) halved
   (c) doubled (d) quadrupled

   Ans: (d) quadrupled

$$\beta = \frac{\lambda D}{d} \ldots (1)$$

$$\beta' = \frac{\lambda\, 2D}{\frac{d}{2}} = \left(\frac{\lambda D}{d}\right) 4 \ldots (2)$$

$$\beta' = 4\beta$$

2. Young's double slit experiment is carried out by using green, red and blue light, one colour at a time. The fringe widths recorded are β_G, β_R, and β_B respectively. Then,
   (d) $\beta_G > \beta_B > \beta_R$ (b) $\beta_B > \beta_G > \beta_R$
   (c) $\beta_R > \beta_B > \beta_G$ (d) $\beta_R > \beta_G > \beta_B$
   Ans: (d) $\beta_R > \beta_G > \beta_B$

$$\lambda_R > \lambda_G > \lambda_B$$

$$\beta_R > \beta_G > \beta_B \text{ as } \beta \propto \lambda$$

3. Which one of the following statements is true?
   (a) both light and sound waves can travel in vacuum
   (b) both light and sound waves in air are transverse
   (c) the sound waves in air are longitudinal, while the light waves are transverse
   (d) both light and sound waves in air are longitudinal
   Ans: (c) the sound waves in air are longitudinal, while the light waves are transverse
4. The frequency of light wave in a material is 2×10^{14} Hz and wavelength is 5000 A^0. The refractive index of material will be
   (a) 1.50 (b) 3.00 (c) 1.33 (d) 1.40
   Ans: (b) 3.00

$$\mu = c/v_m$$

$$= c/\upsilon\lambda$$

$$\mu = \frac{3 \times 10^8}{(2 \times 10^{14})(5000 \times 10^{-10})}$$

$$\mu = \frac{3 \times 10^8}{10^4 \times 10^4}$$

$$\therefore \quad \mu = 3$$

5. Interference was observed in an interference chamber, when air was present. Now the chamber is evacuated and if the same light is used, a careful observer will see
   (a) no interference
   (b) interference with bright bands
   (c) interference with dark bands
   (d) interference, in which width of the fringe will be slightly increased

   Ans: (d) interference, in which width of the fringe will be slightly increased

   When chamber is evacuated, the refractive index (μ) decreases. Therefore, wavelength $\left(\lambda \propto \frac{1}{\mu}\right)$ increases.

   Now, fringe width is given by

   $$\beta = \frac{\lambda D}{d}$$

   $$\Rightarrow \beta \propto \lambda$$

   Therefore, as wavelength increases, the fringe width will increase.

6. Light travels through a glass plate of thickness t and having refractive index μ. If c is the velocity of light in vacuum, the time taken by the light to travel this thickness of glass is
   (a) tμc (b) tc/μ (c) μc/t (d) μt/c

   Ans: (d) μt/c

   The speed of light in glass plate, $v = \frac{c}{\mu}$

   Therefore, time taken by light to traverse the thickness of the glass plate = $\frac{t}{c/\mu} = \frac{\mu t}{c}$

7. A parallel beam of monochromatic light of wavelength 5000 A^o is incident normally on a single narrow slit of width 0.001 mm. The light is focused by a convex lens on a screen placed in focal plane. The first minimum will be formed for the angle of diffraction equal to
   (a) 0° (b) 15° (c) 30° (d) 50°

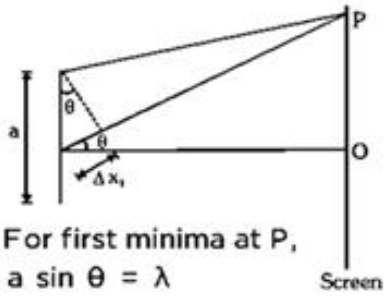
   Ans: (c) 30°

$$a \sin\theta = n\lambda$$
$$\sin\theta = \frac{n\lambda}{a}$$
$$\sin\theta = \frac{1\times5000\times10^{-10}}{0.001\times10^{-3}}$$
$$\sin\theta = \frac{1}{2}$$
$$\theta = \sin^{-1}\left(\frac{1}{2}\right)$$
$$\theta = 30^\circ$$

8. A parallel beam of fast moving electrons is incident normally on a narrow slit. A fluorescent screen is placed at a large distance from the slit. If the speed of the electrons is increased, which of the following statements is correct?
   (a) diffraction pattern is not observed on the screen in the case of electrons
   (b) the angular width of the central maximum of the diffraction pattern will increase
   (c) the angular width of the central maximum will decrease
   (d) the angular width of the central maximum will be unaffected
   Ans: (c) the angular width of the central maximum will decrease

   The diffraction pattern of electron beam will be observed on the screen. If the speed of electrons is increased, de Broglie wavelength associated with electron beam decreases ($\lambda = h / m v$). As angular width of central maximum is proportional to λ, therefore, angular width of central maximum will decrease.

9. At the first minimum adjacent to the central maximum of a single slit diffraction pattern the phase difference between the Huygen's wavelet from the edge of the slit and the wavelet from the midpoint of the slit is
   (a) π/8 radian (b) π/4 radian (c) π/2 radian (d) π radian
   Ans: (d) π radian



The path difference Δx between the wavelet from the edge of the slit and the wavelet from the midpoint of the slit can be expressed as:

$$\Delta x = \frac{a}{2}\sin\theta$$
$$\Delta\phi = \frac{\Delta x}{\lambda}\times 2\pi$$
$$= \frac{(a/2)\sin\theta}{\lambda}\times 2\pi$$
$$\Delta\phi = \frac{\lambda}{2\lambda}\times 2\pi = \pi \text{ radian}$$

10. A beam of light of λ= 600 nm from a distant source falls on a single slit 1 mm wide and the resulting diffraction pattern is observed on a screen 2 m away. The distance between first dark fringes on either side of the central bright fringe is

(a) 1.2 cm (b) 1.2 mm (c) 2.4 cm (d) 2.4 mm

Ans: (d) 2.4 mm

$$y_n = \frac{n\lambda D}{a}$$

For the first dark fringe ($n = 1$):

$$y_1 = \frac{1 \cdot \lambda D}{a}$$

Substituting the known values:

$$y_1 = \frac{1 \cdot (600 \times 10^{-9}\,\text{m}) \cdot (2\,\text{m})}{1 \times 10^{-3}\,\text{m}}$$

$$y_1 = \frac{600 \times 10^{-9} \cdot 2}{1 \times 10^{-3}}$$

$$y_1 = \frac{1200 \times 10^{-9}}{10^{-3}} = 1200 \times 10^{-6}\,\text{m} = 1.2\,\text{mm}$$

$$\text{Distance between dark fringes} = 2y_1 = 2 \times 1.2\,\text{mm} = 2.4\,\text{mm}$$

For questions 11 to 15, two statements are given- one labelled Assertion (A) and other labelled Reason (R). Select the correct answer to these questions from the options as given below.

A. Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
B. Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
C. Assertion is true but Reason is false.
D. Both Assertion and Reason are false.

11. **Assertion:** A thin film such as soap bubble or a thin layer of oil on water show beautiful colours when illuminated by white light.
Reason: It happens due to the interference of light reflected from upper and lower faces of the thin film.
Ans: **A**
Thin films show colors when illuminated by white light because the reflected rays from the upper and the lower surface of the film interfere constructively to give domination of some color for a particular value of thickness at a point.
12. **Assertion:** Diffraction takes place for all types of waves: mechanical or non-mechanical, transverse or longitudinal.
Reason: Diffraction effects are perceptible only if the wavelength of the wave is comparable to the dimensions of the diffracting device.
Ans: B
13. **Assertion:** Interference is not observed if the two coherent slit sources are broad.
Reason: A broad source is equivalent to many narrow slit sources.

Ans: A

Coherent light from two narrow parallel slits will produce an interference pattern on a screen. A broad source is equivalent to a large number of parallel narrow slits, lying side by side. As there are many slits, there will be many overlapping interference patterns may be formed. The overlapping will be to such an extent that all traces of a fringe system will be lost. Thus, almost even illumination will be observed. There will be no interference pattern seen. Thus, for obtaining a distinct interference fringe pattern, a narrow slit is required.

14. **Assertion:** For best contrast between maxima and minima in the interference pattern of Young's double slit experiment, the amplitudes of light waves emerging from the two sources should be equal.
    Reason: For interference, the sources must be coherent.
    Ans: B
15. **Assertion:** When a thin transparent sheet is placed in front of both the slits of Young's experiment, the fringe width will remain the same.
    Reason: In Young's experiment, the fringe width is directly proportional to the wavelength of the source used.
    Ans: B